

AI-Based Restoration of Degraded Images

Team NanoRestore | Vellore Institute of Technology - Chennai

Deenadayalan.B [CAPTAIN]

Role: Team Lead & AI Architecture Specialist

Email: deenadayalanb20081030@gmail.com | Phone: +91 7667407910

Key Contribution: Designed NAFNetSR model with SimpleGate & SCA, multi-scale U-Net encoder-decoder.

Mohamed Faiz.Y

Role: Metrology Loss & Optimization Lead

Email: mf5079549@gmail.com | Phone: +91 9629180309

Key Contribution: Designed Composite Metrology Loss (L1 + SSIM + 2D FFT) for sub-0.8nm LER preservation.

Jai Ganesh.V

Role: GPU Acceleration & Benchmark Lead

Email: vjaiganesh143@gmail.com | Phone: +91 8072604508

Key Contribution: Engineered NVIDIA H100 Tensor Core inference pipeline with dynamic batching & async I/O.

Rohita.K

Role: Data Pipeline & Web Platform Lead

Email: rohita.k2421@gmail.com | Phone: +91 7550068135

Key Contribution: Built 10,000+ synthetic wafer dataset generator, raw .NPY parser/exporter, and web studio.

Problem Statement: AI-Based Restoration of Degraded Images

Physical Degradation Mechanisms in Semiconductor Metrology & Defect Inspection

1. Multiplicative Speckle Noise

Coherent laser illumination in optical wafer inspection creates severe multiplicative Gamma/Rayleigh speckle noise.

Key Physical Challenge: Speckle interference causes constructive pixel intensity peaks to exceed ground truth bounds (> 1.0 , up to $1.6+$). Naive clipping ruins restoration fidelity.

2. Gaussian Sensor & Thermal Noise

High-speed line-scan optical sensors and low-dose scanning electron microscopy (SEM) introduce Gaussian thermal and Poisson shot noise.

Industrial Impact: Thermal noise buries sub-5nm defects (bridging, dendrite crystal growth, open vias) on 3nm/5nm logic and 3D NAND memory dies.

3. 2x Spatial Resolution Loss

Optical point spread function (PSF) diffraction and fab scanning speed trade-offs result in downsampled low-resolution captures (128×128 & 256×256).

Goal: Jointly perform speckle denoising and 2x super-resolution ($128 \rightarrow 256$ & $256 \rightarrow 512$) while preserving line-edge roughness ($dLER < 0.8nm$).

Idea Description: NAFNetSR Metrology Architecture

Nonlinear Activation Free Super-Resolution & Joint Speckle Denoising

Why NAFNetSR for Semiconductor Fabs?

1. Nonlinear Activation Free (NAF): Eliminates expensive GELU/ReLU activations via SimpleGate ($x1 * x2$), maximizing GPU Tensor Core arithmetic intensity.

2. Simplified Channel Attention (SCA) Uses $2D$ Fast Fourier Transform (FFT) to prevent overfitting on periodic lithography patterns (AvgPool + 1×1 Conv).

* Dynamic Resolution Immunity: Auto-pads arbitrary dimensions to multiple scales. Utilizes Topology-aware residual learning to preserve edge gradients across decoder stages with skip connections.

4. Universal Data Support: Natively reconstructs 30 hr frequency details ($128 \times 256 \times 256$) with zero checkerboard artifacts.

Joint Tri-Modal Degradation Handling

* Unclipped Speckle Pipeline: Processes unclipped raw float intensities (> 1.0) and strictly clamps restored predictions to $[0.0, 1.0]$ only at the output layer.

A. Synthetic Degradation Pipeline

- Multiplicative Gamma Speckle: $\eta \sim \text{Gamma}(k=10, \theta=0.1)$ yielding unclipped >1.0 float values
- Additive Gaussian Thermal Noise: σ in $[0.01, 0.03]$
- Poisson Electron Shot Noise: peak in $[80, 200]$
- 2x Bicubic Spatial Downsampling (512 $\rightarrow$ 256 & 256 $\rightarrow$ 128)

B. NAFNetSR Architecture & Head

- Input Layer: Dynamic channel adaptation (Grayscale/RGB) + Dynamic Reflection Pad
- Encoder-Decoder: 4 Stages $[c=32, 64, 128, 256]$ with SimpleGate + SCA
- Residual Skip Connections: Direct gradient propagation across scales
- Upsampler: 1×1 Conv + PixelShuffle(scale=2) + Dynamic Unpad

C. Composite Metrology Loss (Sub-0.8nm LER)

$$L_{\text{total}} = L_{\text{L1}} + 0.5 * L_{\text{SSIM}} + 0.05 * L_{\text{FFT}}$$

- L_{L1} (Charbonnier): Robust pixel fidelity against speckle outliers
- L_{SSIM} : Structural similarity and local pattern contrast
- L_{FFT} : Frequency-domain 2D Fourier loss for sharp edge & periodic line retention

Innovation & Uniqueness: What Sets Our Solution Apart

Engineering Breakthroughs for Real-World Fab Deployment

01. Pure Floating-Point Dynamic Range Physics

Correctly handles laser speckle intensities exceeding 1.0 without premature clipping, preserving true constructive interference dynamics and preventing severe non-linear distortion.

02. Ultra-High H100 GPU Throughput (~15,789 Tiles/Min)

Engineered with torch.inference_mode, mixed precision (FP16/BF16), and multi-threaded async disk I/O, achieving 3.8ms latency per full wafer tile on NVIDIA H100.

03. Universal 8-Modality Semiconductor Coverage

10,000+ sample suite covering Logic FinFET (3nm), 3D NAND, DRAM, TSVs, EUV Gratings, CMP Scratches, SEM Dendrites, and Out-of-Distribution multi-material samples.

04. Live Interactive Web Studio with Pure JS .NPY Engine

Zero-backend browser platform with custom DataView binary .NPY parser, real-time split-wipe viewer, and residual error heatmap with sub-pixel probing.

10,000-Wafer Tile Multi-Model Metrology Benchmark (NVIDIA H100 SXM5)

Model Architecture	PSNR (dB)	SSIM	dLER (nm)	Latency	Throughput	Fab Tiles/Min	VRAM

* NAFNet-Metrology (Ours)	20.99 dB	0.8322	< 0.29 nm	3.8 ms	263.2 FPS	~15,789/min	< 2.4 GB
Restormer-Lite (Transformer)	20.87 dB	0.8278	0.28 nm	7.4 ms	135.1 FPS	~ 8,108/min	4.6 GB
SwinIR-Metrology (Swin-T)	20.80 dB	0.8252	0.28 nm	9.1 ms	109.9 FPS	~ 6,593/min	5.8 GB
UNet-Baseline (Standard)	20.40 dB	0.8141	0.28 nm	4.5 ms	222.2 FPS	~13,333/min	2.8 GB

Key Quantitative Takeaways:

- NAFNet-Metrology achieves the highest PSNR & SSIM with 2x faster throughput than Vision Transformers.
- Preserves critical dimension line-edge roughness with dLER < 0.29 nm (Target: < 0.8 nm).
- Multi-modal test sets across all 8 semiconductor industries restored with 100% success rate.

Technology Stack & Industrial Fab Feasibility

Software Architecture, Computational Footprint & Deployment Readiness

Software Framework

PyTorch 2.2+, TorchScript / CUDA Graph, Python 3.11, NumPy, OpenCV, PIL

Target Hardware

NVIDIA H100 (80GB SXM5) / RTX 4090 / Cloud Tensor Cores

Model Footprint

1.2 Million Parameters (~4.8 MB Checkpoint), Peak VRAM < 2.4 GB

Training Efficiency

~25 minutes for 50 epochs on H100 with Mixed Precision (BF16/FP16)

Inference Latency

3.8 ms / tile (~263 FPS) -> ~15,789 Wafer Reticle Tiles / minute

Zero-Downtime Fallback

Integrated high-speed Vectorized NumPy engine for pure CPU environments

Official Public GitHub Repository

Repository URL:

<https://github.com/deenadayalanb20081030-cloud/KLA-AI-Image-Restoration>

Includes All Mandatory Submission Components:

- [OK] README.md (Comprehensive setup & 1-line execution)
- [OK] evaluate.py (Standalone CLI benchmark accepting .npy & images)
- [OK] train.py (Reproducible supervised training script)
- [OK] weights/best_model_weights.pt (Final trained PyTorch model)
- [OK] outputs/ (1,000+ restored full-res PNGs & raw .npy arrays)
- [OK] requirements.txt (Pinned dependencies)

Live Interactive Metrology Studio

Live Demo URL:

<https://deenadayalanb20081030-cloud.github.io/KLA-AI-Image-Restoration/>

Interactive Features for Reviewers:

- [OK] Real-time interactive Wipe Split-Slider (512x512)
- [OK] 3-Way Grid comparison (Ground Truth vs NoisyLR vs Restored)
- [OK] Residual Error Heatmap with sub-pixel delta probe
- [OK] Pure JavaScript Binary .NPY Array Parser & Exporter
- [OK] 8 Semiconductor Fab Modality Selectors
- [OK] Real-time Loss Function & Pareto Frontier Workshop

Academic Papers & Industrial Standards

1. Chen, L., Chu, X., Zhang, X., & Sun, J. (2022). Simple Baselines for Image Restoration (NAFNet). European Conference on Computer Vision (ECCV 2022).
2. Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., & Yang, M. H. (2022). Restormer: Efficient Transformer for High-Resolution Image Restoration. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2022).
3. Liang, J., Cao, J., Sun, G., Zhang, K., Van Gool, L., & Timofte, R. (2021). SwinIR: Image Restoration Using Swin Transformer. IEEE/CVF International Conference on Computer Vision Workshops (ICCVW 2021).
4. Goodman, J. W. (2007). Speckle Phenomena in Optics: Theory and Applications. Roberts & Company Publishers. (Coherent laser speckle physics in optical wafer metrology).
5. SEMI Metrology Standards (2024). Standard Guide for Scanning Electron Microscope Defect Review and Critical Dimension (CD-SEM) Inspection in Semiconductor Manufacturing.